



## Koneru Lakshmaiah Education Foundation

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### STUDENT CASE STUDY EXECUTION REPORT

|                |                             |
|----------------|-----------------------------|
| Department     | FED                         |
| Subject        | DSA -2                      |
| Academic year  | I – III Semester(2025-26)   |
| Faculty Name   | Mohammed Razia Alangir Banu |
| Student Name   | B.Deepak Goud               |
| Roll Number    | 2520080035                  |
| Branch         | AI&DS                       |
| Course Outcome | 5                           |

#### Case Study Topic : StreamFlix Analytics — sorting massive watch-history logs

A StreamFlix analytics cluster processes **60 GB of user watch-history records** every night.

#### Word Done Summary:

The **StreamFlix Analytics** project was developed to efficiently process and analyze large-scale watch-history data generated by users on the streaming platform. The system handles approximately **60 GB of watch-history records every night**, ensuring that valuable viewing insights can be extracted from massive datasets.

To manage the large volume of data, advanced sorting algorithms were implemented to organize watch-history logs based on factors such as viewing time, user activity, and content popularity. Efficient sorting significantly reduced data processing time and improved overall system performance.

The analytics pipeline was designed to identify viewing trends, popular content categories, and user engagement patterns. By processing sorted records, the system generates meaningful reports that help understand audience behavior and content consumption habits.

Overall, the project successfully demonstrated the application of efficient data sorting techniques in big-data environments. It improved data organization, accelerated analytics operations, and enabled faster decision-making through accurate and well-structured watch-history analysis.

### Implementation Details:

**Data Collection** – Imported 60 GB of watch-history logs from the StreamFlix database.

**Data Preprocessing** – Cleaned and formatted records for efficient processing.

**Sorting Implementation** – Applied efficient sorting algorithms to organize large datasets.

**Analytics Processing** – Analyzed sorted data to identify viewing patterns and trends.

**Report Generation** – Generated insights and summaries for decision-making and performance monitoring.

OutPut:

```
● b.deepak@BDeepaks-MacBook-Air DSA 2 % javac MergeSortDemo.java
● b.deepak@BDeepaks-MacBook-Air DSA 2 % java MergeSortDemo
Original Array:
[10, 4, 15, 2, 13, 6, 11, 1, 9, 5, 14, 8]

Sorted Array:
[1, 2, 4, 5, 6, 8, 9, 10, 11, 13, 14, 15]
○ b.deepak@BDeepaks-MacBook-Air DSA 2 % █
```

GitHub Link:<https://github.com/bdeepakg-boop/DSA-Project/upload/main>

| Student Signature | Faculty Signature |
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